

1. Perceptron learning algorithm (supervised learning algorithm):

A. supervised learning: when training a model (neuron network), we will offer not only "inputs x_i " but also corresponding "output y (ground truth)" as pair training example [inputs x_i , output y (ground truth)] for the model to learn.

B. processes of perceptron learning algorithm: step1. randomly initialize the weights w_i and given pair training example [inputs x_i , output y (ground truth)]
step2. select one pair training example [inputs x_i] for perceptron to compute corresponding "predicted output P_y "
step3. match predicted output P_y & output y (ground truth):

if "($p_y < 0$) != ($y = +1$)" = " ($p_y = -1$) != ($y = +1$)" ,then " add each $w[i]$ by $\eta * x[i]$ " = " $w[i] += (y * \eta * x[i]), y = +1$ "

if " ($p_y > 0$) != ($y = -1$)" = " ($p_y = +1$) != ($y = -1$)" ,then " subtract each $w[i]$ by $\eta * x[i]$ " = " $w[i] += (y * \eta * x[i]), y = -1$ "

step4. repeat "step2 & step3" until predicted output $P_y =$ output y (ground truth).

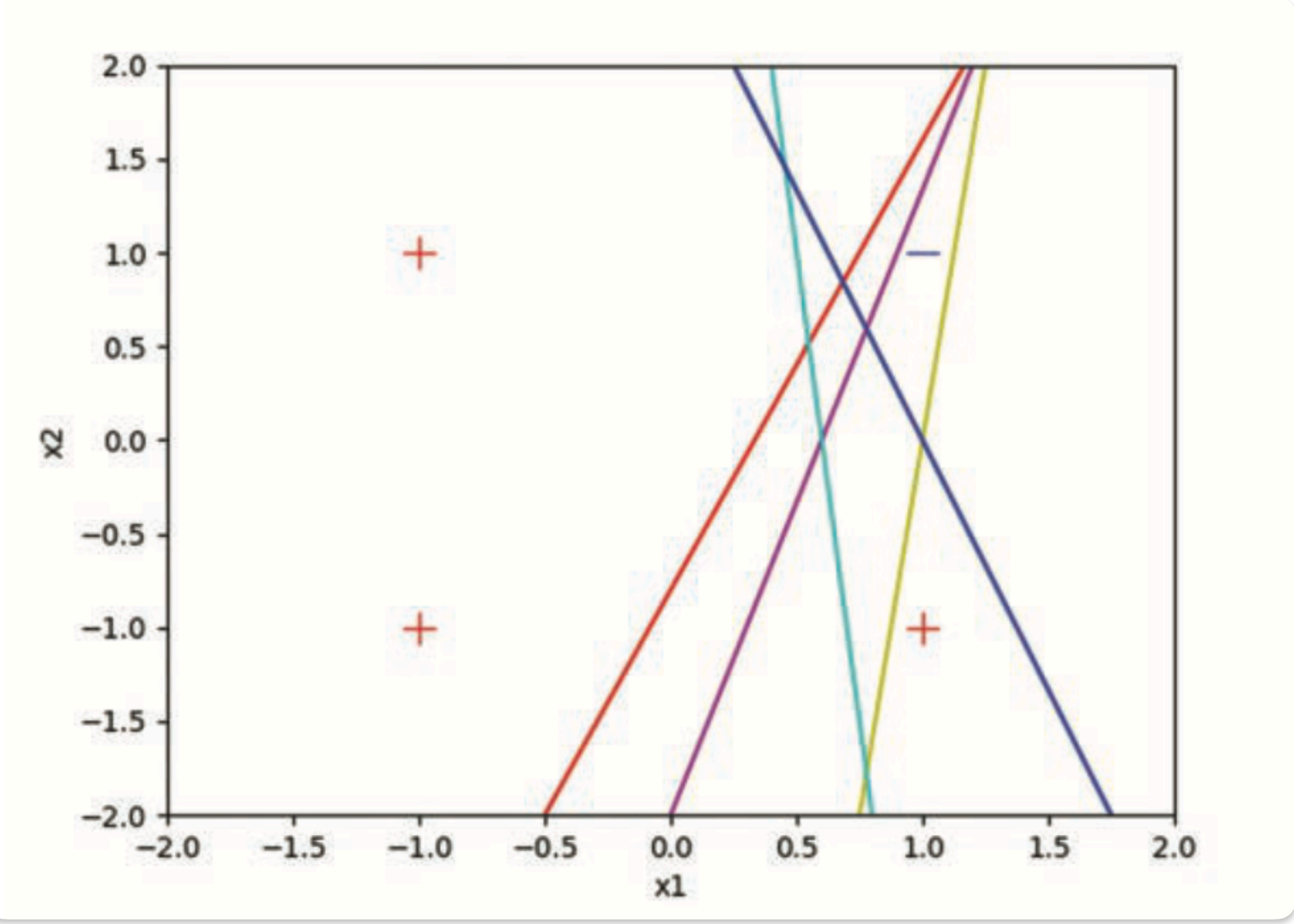
2. Learning rate (η): an example of hyperparameter! which is not a parameter adjusted by perceptron learning algorithm but by user.

3. Decision boundary equation:

A. In two-input perceptron, there is a line ($w_0x_0 + w_1x_1 + w_2x_2 = 0$) in 2D graph, the line can divide the graph into 3 parts. $\Rightarrow$ Part1. on the line: $w_0x_0 + w_1x_1 + w_2x_2 = 0$
Part2. one side of the line: $w_0x_0 + w_1x_1 + w_2x_2 > 0$
Part3. the other side of the line: $w_0x_0 + w_1x_1 + w_2x_2 < 0$

B. the line: $w_0x_0 + w_1x_1 + w_2x_2 = 0 \rightarrow x_2 = -\frac{w_1}{w_2}x_1 - \frac{w_0}{w_2}$
 (Annotations: slope of line points to $-\frac{w_1}{w_2}$, y-intercept of line points to $-\frac{w_0}{w_2}$)

C.



$\Rightarrow$ the red line (built by initial weights) cannot divide the graph (one side for output= $\oplus$, the other side for output= $\ominus$)

$\Rightarrow$ purple line

$\Rightarrow$ yellow line

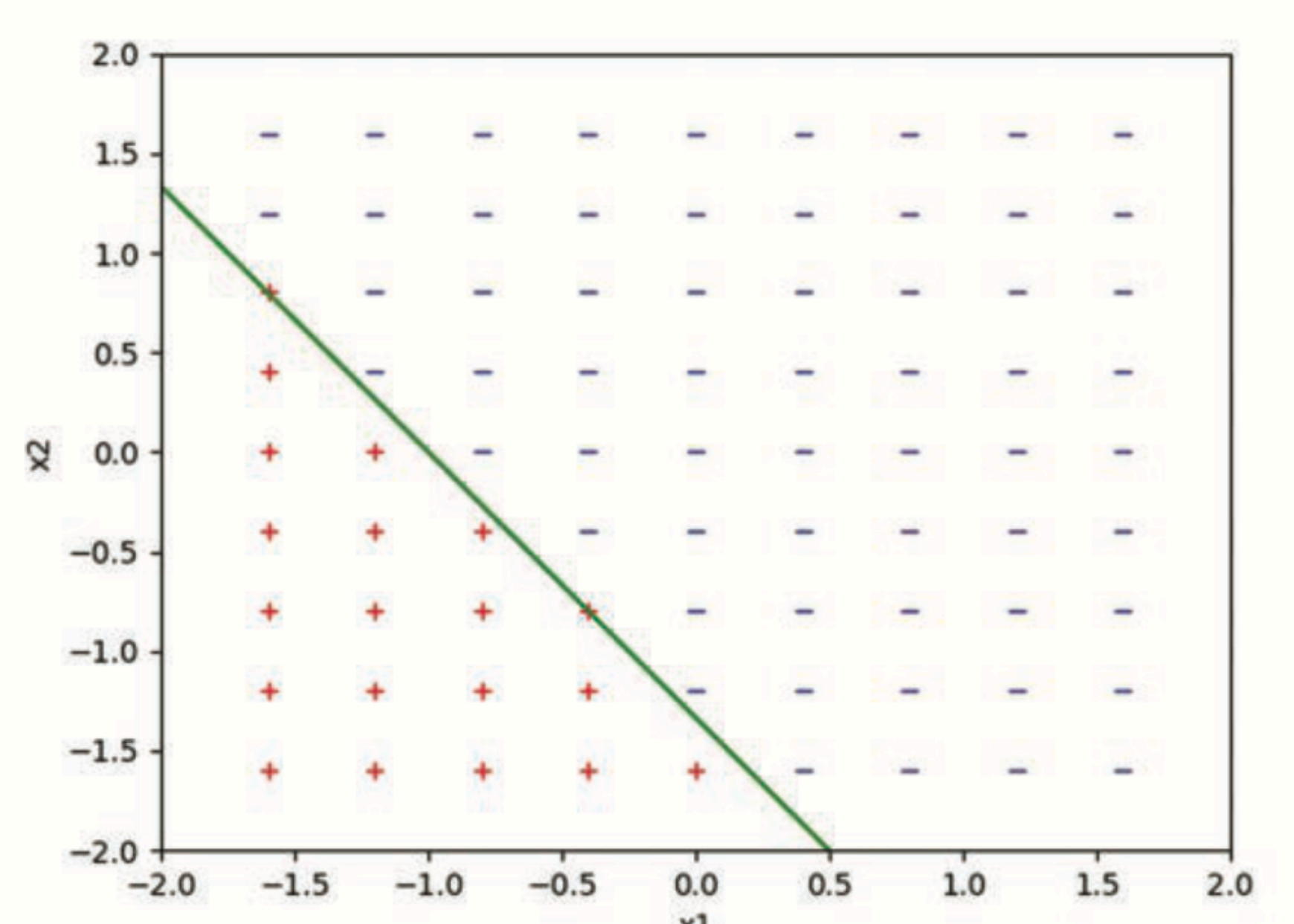
$\Rightarrow$ bright blue line

$\Rightarrow$ dark blue line (built by finally adjusted weight) can divide the graph (one side for output= $\oplus$, the other side for output= $\ominus$)

$\Rightarrow$ all predicted output $P_y =$ output y (ground truth), and perceptron learning algorithm converge,

$\Rightarrow$ the dark blue line can divide the graph into 3 parts. $\Rightarrow$ Part1. on the line: $w_0x_0 + w_1x_1 + w_2x_2 = 0$
Part2. one side of the line (the side must be the area full of output= $\oplus$): $w_0x_0 + w_1x_1 + w_2x_2 > 0$
Part3. the other side of the line (the side must be the area full of output= $\ominus$): $w_0x_0 + w_1x_1 + w_2x_2 < 0$

D.



$\Rightarrow$ all predicted output $P_y =$ output y (ground truth), and perceptron learning algorithm converge

$\Rightarrow$ the green line can divide the graph into 3 parts. $\Rightarrow$ Part1. on the line: $w_0x_0 + w_1x_1 + w_2x_2 = 0$
Part2. one side of the line (the side must be the area full of output= $\oplus$): $w_0x_0 + w_1x_1 + w_2x_2 > 0$
Part3. the other side of the line (the side must be the area full of output= $\ominus$): $w_0x_0 + w_1x_1 + w_2x_2 < 0$

Figure 1-5 Output of a perceptron as a function of two inputs x_1 and x_2